**Department of Statistics & Computer Science**
**University of Kelaniya**
**ACADEMIC YEAR-2023/2024**
**COSC 21063-Data Structures and Algorithms**
**Practical Tutorial 09**

**1.** The following table contains the sales information of the UBuy online store for last Friday.

| **Product ID** | **Product Name** | **Category** | **Unit Price (Rs.)** | **Quantity Sold on Friday** |
|---|---|---|---|---|
| P108 | Wireless Mouse | Electronics | 2,160 | 30 |
| P034 | Handbag | Accessories | 3,450 | 5 |
| P078 | Phone cover | Accessories | 1,750 | 26 |
| P105 | Bluetooth Speaker | Electronics | 13,780 | 5 |
| P003 | Ladies blouse | Clothing | 1,650 | 18 |
| P053 | Shampoo | Groceries | 2,370 | 20 |
| P114 | Laptop | Electronics | 250,000 | 3 |
| P004 | Frock | Clothing | 4,520 | 10 |
| P117 | Earphone | Electronics | 7,860 | 2 |
| P120 | Microwave oven | Electronics | 56,830 | 15 |

a. Using contiguous list data structure input above sales details.
b. Calculate the sales amount of each product on last Friday and display them.
c. Sort the sales details according to the sales amount of each product using selection sort and display them.
d. There is a mistake in the quantity sold in frocks. It should be corrected as 49. Make the correction and display the updated list of products with their sales amount.
e. Find and display the products whose total sale amount is greater than Rs. 40,000.
f. Identify products that belong to the Electronics category and calculate the sales amount received from electronics category.
g. Calculate the percentage of sale amount of electronic category relative to the total sale amount of all categories.

**2.**

| **Product ID** | **Product Name** | **Category** | **Unit Price (Rs.)** | **Quantity Sold** |
|---|---|---|---|---|
| T101 | Smartphone | Electronics | 75,000 | 15 |
| T102 | Smartwatch | Electronics | 25,000 | 40 |
| T103 | Wireless Charger | Accessories | 2,500 | 60 |
| T104 | Laptop | Electronics | 120,000 | 10 |
| T105 | Laptop Bag | Accessories | 3,000 | 25 |
| T106 | USB Cable | Accessories | 700 | 100 |
| T107 | Bluetooth Speaker | Electronics | 10,000 | 18 |
| T108 | Headphones | Electronics | 5,000 | 30 |
| T109 | Mouse Pad | Accessories | 900 | 50 |
| T110 | Tablet | Electronics | 45,000 | 12 |

a. Input the above sales details into a contiguous list data structure.
b. Calculate and display the total sales amount (Unit Price × Quantity Sold) for each product.
c. Sort the list by total sales amount in descending order using Bubble Sort and display the sorted list.
d. Sort the list by product name in ascending alphabetical order using Insertion Sort and display the sorted list.
e. Sort the list by quantity sold in ascending order using Selection Sort and display the sorted list.
f. Using Sequential Search, find the product with the name "USB Cable" and update its quantity sold to 120. Then display the updated product details.
g. Find and display all products with a total sales amount greater than Rs. 50,000.
h. Identify all products in the Accessories category, calculate the total sales amount for this category, and compute the percentage of this amount relative to the total sales amount of all products.

**3.** The following table contains information about bank customers in ABC bank.

| Account Number | Customer Name | Account Type | Account Balance (as of 2024.01.01) |
|---|---|---|---|
| 1001 | Kamal Dissanayake | savings | Rs. 500 000 |
| 1002 | Namal Perera | current | Rs. 975 000 |
| 1003 | Sithumm Udovita | current | Rs. 100 000 |
| 1004 | Manel Dias | savings | Rs. 1 250 000 |
| 1005 | Chethiya Munasinghe | savings | Rs. 950 000 |
| 1006 | Sanju Perera | current | Rs. 1 500 000 |
| 1007 | Lahiru Karunarathna | savings | Rs. 600 000 |
| 1008 | Tharanga Prasad | savings | Rs. 400 000 |
| 1009 | Shashi Dayarathna | savings | Rs. 250 000 |
| 1010 | Anju Senanayake | current | Rs. 1 100 000 |

a. Using contiguous list data structure input above details and display them.
b. Calculate the interest amount earned by the customer based on their account balance and following bank interest rates.

**Interest Amount = Account Balance × Annual Interest Rate**

c. Account Balance below Rs.250 000 - Annual Interest Rate = 2.5%
d. Account Balance between Rs.250 000 and Rs.500 000 - Annual Interest Rate = 5.0%
e. Account Balance between Rs.500 000 and Rs.750 000 - Annual Interest Rate = 7.5%
f. Account Balance between Rs.750 000 and Rs.100 0000 - Annual Interest Rate = 8.0%
g. Account Balance between Rs.1 000 000 and Rs.1 500 000 - Annual Interest Rate = 9.5%
h. Calculate the Total Account Balance for each customer after one year.
**Hint:**

**Total Account Balance = Account Balance + Interest Amount**

i. Sort the customer details according to Total Account Balance using quick sort and display them.
j. Find and display all customers who have more than 1 000 000 in their account after one-year period.
k. Identify customers with savings accounts and calculate the total balance in savings accounts.
l. Calculate the percentage of the total balance in savings accounts relative to the total balance in all accounts.